

TPG4190 Seismic data acquisition and processing

Lecture 5: Deghosting

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Overview

- ▶ The wave equation
- ▶ Up- and downgoing waves
- ▶ Removal of the ghost effect

The acoustic wave equation

The acoustic wave equation describes wave motion in a material with no shear forces, i.e. only P-waves can exist.

$$\rho(\mathbf{x})\ddot{u}_i(\mathbf{x}, t) = \partial_i \sigma(\mathbf{x}, t) + f_i(\mathbf{x}, t), \quad (1)$$

$$\sigma(\mathbf{x}, t) = \kappa(\mathbf{x})\partial_i u_i(\mathbf{x}, t) + q(\mathbf{x}, t). \quad (2)$$

- ▶ ρ : Density
- ▶ u_i : Component i of particle displacement
- ▶ σ : $-p$ where p is the pressure.
- ▶ κ : Bulk modulus
- ▶ f_i : component i of source body force
- ▶ q : source of volume injection type

The acoustic wave equation

Divide equation (2) with density and differentiate equation (2) two times w.r.t. time:

$$\ddot{u}_i(\mathbf{x}, t) = \frac{1}{\rho(\mathbf{x})} \partial_i \sigma(\mathbf{x}, t) + \frac{f_i(\mathbf{x}, t)}{\rho(\mathbf{x})}, \quad (3)$$

$$\ddot{\sigma}(\mathbf{x}, t) = \kappa(\mathbf{x}) \partial_i \ddot{u}_i(\mathbf{x}, t) + \ddot{q}(\mathbf{x}, t). \quad (4)$$

Insert equation (3) into equation (4) to obtain:

$$\ddot{\sigma}(\mathbf{x}, t) = \kappa(\mathbf{x}) \partial_i \left[\frac{1}{\rho(\mathbf{x})} \partial_i \sigma(\mathbf{x}, t) \right] + \partial_i \left[\frac{f_i(\mathbf{x}, t)}{\rho(\mathbf{x})} \right] + \ddot{q}(\mathbf{x}, t). \quad (5)$$

The acoustic wave equation

Assume $f = 0$ and constant density $\rho = \rho_0$

$$\ddot{\sigma}(\mathbf{x}, t) = c^2(\mathbf{x}) \partial_i \partial_i \sigma(\mathbf{x}, t) + \ddot{q}(\mathbf{x}, t). \quad (6)$$

where $c(\mathbf{x}) = \sqrt{\kappa(\mathbf{x})/\rho_0}$ or

$$\nabla^2 \sigma(\mathbf{x}, t) - \frac{1}{c^2(\mathbf{x})} \ddot{\sigma} = s(\mathbf{x}, t) \quad (7)$$

where $s(\mathbf{x}, t) = -\ddot{q}(\mathbf{x}, t)/c^2(\mathbf{x})$

The acoustic wave equation

We now assume that the velocity is independent of x and y coordinates and only depends on depth; $c(\mathbf{x}) = c(z)$

$$\nabla^2 \sigma(\mathbf{x}, t) - \frac{1}{c^2(z)} \ddot{\sigma} = s(\mathbf{x}, t) \quad (8)$$

∇^2 operator is

$$\nabla^2 \sigma(\mathbf{x}, t) = \frac{\partial^2 \sigma}{\partial_x^2}(\mathbf{x}, t) + \frac{\partial^2 \sigma}{\partial_y^2}(\mathbf{x}, t) + \frac{\partial^2 \sigma}{\partial_z^2}(\mathbf{x}, t) \quad (9)$$

The Fourier transform over x - and y is obtained by replacing $\partial_x^2 \rightarrow (-ik_x)^2 = -k_x^2$ and $\partial_y^2 \rightarrow (-ik_y)^2 = -k_y^2$.

The acoustic wave equation

$$\nabla^2 \sigma(\mathbf{x}, t) \rightarrow (-k_x^2 - k_y^2) \sigma(k_x, k_y, z, t) + \frac{\partial^2 \sigma}{\partial z^2}(k_x, k_y, z, t) \quad (10)$$

We also have for the fourier transform over time

$$\frac{\partial^2 \sigma}{\partial t^2}(\mathbf{x}, t) \rightarrow -\omega^2 \sigma(k_x, k_y, \omega) \quad (11)$$

Putting this together we get for the wave equation

$$-(k_x^2 + k_y^2) \sigma(\mathbf{x}, t) + \frac{d^2 \sigma}{dz^2}(k_x, k_y, z, \omega) + \frac{\omega^2}{c^2(z)} \sigma(k_x, k_y, z, \omega) \quad (12)$$

$$= s(k_x, k_y, z, \omega) \quad (13)$$

The acoustic wave equation

Using

$$k_z^2 = \omega^2/c^2(z) - (k_x^2 + k_y^2) = \omega^2/c^2(z) - k_r^2, \quad (14)$$

where $k_r^2 = k_x^2 + k_y^2$.

we have

$$\frac{d^2\sigma(k_r, z, \omega)}{dz^2} + k_z^2(k_r, z, \omega)\sigma(k_r, z, \omega) = s(k_r, z, \omega) \quad (15)$$

The acoustic wave equation

The solution of the this wave equation is

$$\sigma(k_r, z, \omega) = U(k_r, \omega) \exp(ik_z z) + D(k_r, \omega) \exp(-ik_z z), \quad (16)$$

Where the constants U and D are interpreted as the amplitude of the upgoing wave and the downgoing wave, respectively.

We also have from the equation of motion

$$\rho_0 \ddot{u}_z(\mathbf{x}, t) = \rho_0 \dot{v}_z(\mathbf{x}, t) = \frac{\partial \sigma(\mathbf{x}, t)}{\partial z}. \quad (17)$$

Or in the wave-number frequency domain

$$-i\omega \rho_0 v_z(k_r, \omega, z) = \frac{\partial \sigma(k_r, \omega, z)}{\partial z}, \quad (18)$$

$$v_z(k_r, \omega, z) = \left(\frac{-1}{i\omega \rho_o} \right) \frac{\partial \sigma(k_r, \omega, z)}{\partial z}. \quad (19)$$

The acoustic wave equation

Using the last equation together with equation (16)

$$v_z(k_r, \omega, z) = \left(\frac{-ik_z}{i\omega\rho_o} \right) [U(k_r, \omega) \exp(ik_z z) - D(k_r, \omega) \exp(-ik_z z)], \quad (20)$$

For simplicity we set $z = 0$ at the receiver depth. Equation (16) and (20) then gives

$$\left(\frac{i\omega\rho}{-ik_z} \right) v_z(k_r, \omega) = U(k_r, \omega) - D(k_r, \omega), \quad (21)$$

$$\sigma(k_r, \omega) = U(k_r, \omega) + D(k_r, \omega). \quad (22)$$

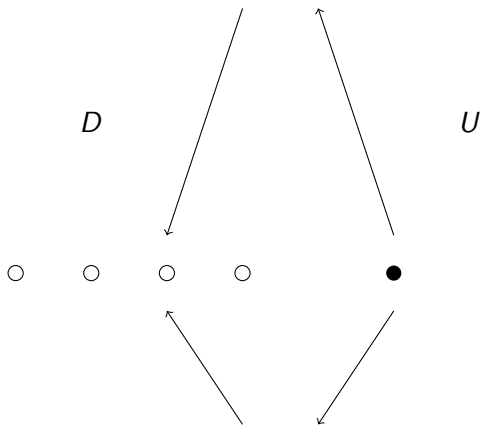
U-D separation

Adding and subtracting equations (22) and (22) gives

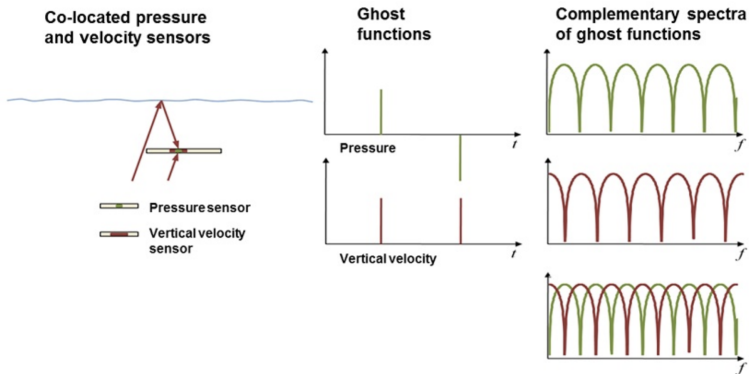
$$U(k_r, \omega) = \left(\frac{i\omega\rho}{-2ik_z} \right) v_z(k_r, \omega) + \sigma(k_r, \omega), \quad (23)$$

$$D(k_r, \omega) = - \left(\frac{i\omega\rho}{-2ik_z} \right) v_z(k_r, \omega) + \sigma(k_r, \omega). \quad (24)$$

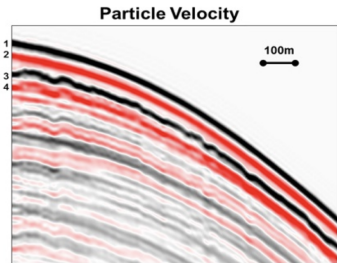
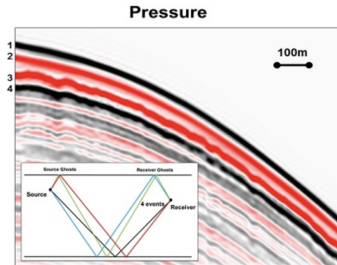
U-D deghosting



U-D deghosting

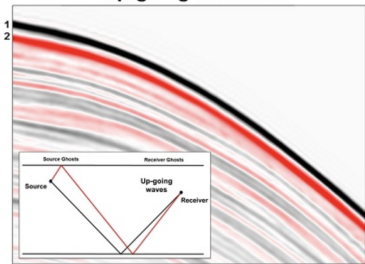


U-D deghosting



U-D deghosting

Up-going Wavefield



Down-going Wavefield

